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POSITION STATEMENT

Diagnostic approaches for infants with cholestatic liver diseases: Position paper and perspectives of the Federation of International Societies of Pediatric Gastroenterology, Hepatology, and Nutrition

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Abstract

Cholestasis in infancy is the most common manifestation of liver disease in children, with some patients progressing to cirrhosis or liver failure necessitating transplantation. Neonatal cholestasis remains a diagnostic challenge, as it requires differentiation of cholestatic infants from a large number of jaundiced newborns with benign causes. The first step is to diagnose patients with biliary atresia (BA) as early as possible to ensure timely surgery-Kasai portoenterostomy (KPE). Universal newborn screening using stool color cards or direct bilirubin measurements have been shown to identify patients before the onset of symptoms. Multiple diagnostic modalities, including clinical history, physical examination, laboratory tests, emerging biomarkers, imaging studies, and liver histopathology, can facilitate the decision for intraoperative cholangiography and potential corrective surgery. Advances in diagnostic testing, particularly genetic sequencing, have greatly enhanced our ability to evaluate and manage infants with cholestasis. Given highly variable resources and access to these new diagnostic modalities, local flexibility and adaptability should be implemented within each institution and medical care system to foster seamless collaboration between primary care physicians and specialized centers with expertise in genetic diagnosis, KPE, and liver transplantation. This report provides updates on the evaluation of neonatal cholestasis, including insights into screening, diagnosis, and genetic testing, along with future perspectives.

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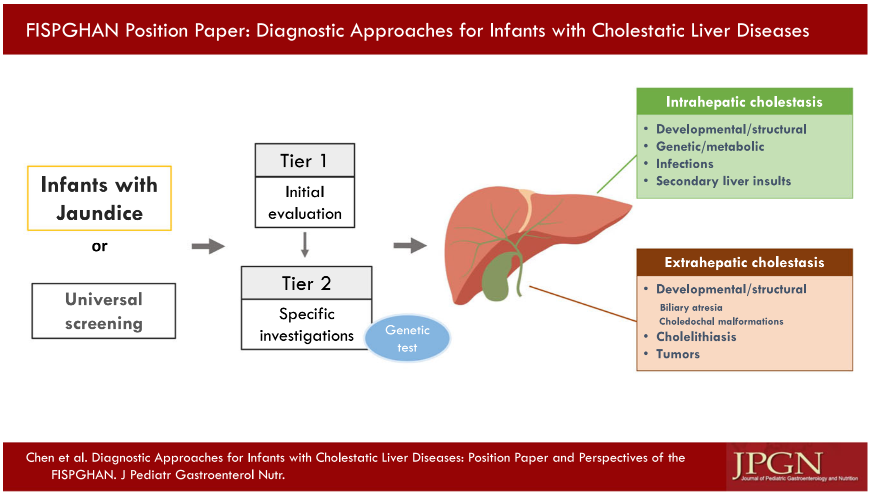

KEYWORDS

Alagille syndrome, biliary atresia, jaundice screening, neonatal cholestasis, progressive familial intrahepatic cholestasis

1 | INTRODUCTION

1.1 | Definition

Cholestasis, broadly defined as impaired bile flow, is the most common manifestation of hepatobiliary disease in infancy. It is estimated to affect 1 in 2500 term infants[1] and is associated with significant morbidity and mortality.[2] Patients with cholestasis may present with jaundice in early infancy, and may manifest hepatomegaly, acholic stools, steatorrhea, dark urine, poor weight gain, coagulopathy, or signs of portal hypertension. As highlighted in the North American and European Societies of Pediatric Gastroenterology, Hepatology and Nutrition (NASPGHAN/ESPGHAN) recommendations, in current practice, cholestasis is defined biochemically as serum direct/conjugated bilirubin levels >1.0 mg/dL (17 μmol/L) in the presence of elevated total bilirubin.[3] An alternative threshold of direct/total bilirubin ratio >20%, though widely applied, may cause confusion in clinical settings; therefore, it is recommended to be used only when the direct/conjugated bilirubin level is equivocal. Rarely, the patients with cholestasis may not present with elevated direct/conjugated bilirubin and can be identified by abnormal serum liver tests, including aspartate aminotransferase (AST), alanine aminotransferase (ALT), gamma-glutamyl transpeptidase (GGT), and bile acids, the latter being a more specific marker for cholestasis with a normal value < 10 μmol/L.[4–6]

1.2 | Etiologies

The etiology of cholestasis is diverse, including extrahepatic cholestasis such as biliary atresia (BA) and choledochal malformation, and intrahepatic cholestasis such as genetic intrahepatic cholestasis and metabolic disorders.[3,4,7] (Figure 1) BA is the most common cause of extrahepatic cholestasis in infants and the leading indication for pediatric liver transplantation (LT) worldwide. The disease progression involves both extrahepatic and intrahepatic fibroinflammatory processes. Early diagnosis and timely Kasai portoenterostomy (KPE) are crucial for optimal outcomes. There are geographical variations in the incidence of BA; it is more common in Asian countries and French Polynesia (1–3/10,000 live births) than in Europe and North America (~0.5/10,000 live births).[8–10] The incidence of other forms of extrahepatic cholestasis, such as choledochal malformations, is less well established.[11]

Among intrahepatic cholestasis, transient neonatal cholestasis (TNC) is common and may be associated with various perinatal factors, including prematurity, infection, perinatal distress, parenteral nutrition (PN), milder forms of genetic and metabolic diseases, or unidentified causes.[12–14] PN-associated cholestasis can occur in the patients receiving PN for 2 weeks or more, particularly in those born prematurely or completely off enteral feeds, and may evolve to intestinal failure-associated liver disease.[15–17] A variety of infectious insults, such as cytomegalovirus (CMV) or urinary tract infections, and metabolic/endocrine disorders such as pan-hypopituitarism, could also lead to cholestasis. Neonatal intrahepatic cholestasis caused by citrin deficiency (NICCD) is more common in the Asian population.[18] In the last two decades, advances in genomic medicine have revealed an expanding list of monogenic liver disorders, many of which had been previously termed "idiopathic neonatal hepatitis" in earlier studies.[3,7,19–26]

1.3 | Unmet needs

Early detection of BA and timely Kasai procedure are key goals in the diagnosis of cholestatic infants and have been advocated by pediatric hepatologists for more than 30 years. However, late referral and delayed Kasai operations still remain significant challenges in real-world settings. A recent survey by the Quality-of-Care Task Force of ESPGHAN revealed a mean age of referral and Kasai surgery at 55 and 61 days, respectively.[27] The majority of regions worldwide do not report the age at Kasai, and presumably, a significant number of patients still experience delayed diagnosis and treatment.

Second, progress in molecular biology has led to the identification of many genetic causes of cholestasis through the development of powerful diagnostic tools, such as next-generation sequencing (NGS)-based genetic analysis. However, the required time, cost, limited availability, and lack of integration of these tools into institutional diagnostic algorithms remain major limitations. Consequently, the genetic testing and disease-specific treatments remain out of reach for many patients.

Global perspective

- Late referral and the Kasai operation remain significant challenges in real-world settings.
- Diagnostic modalities for cholestatic and metabolic diseases, including genetic testing and disease-specific treatments, still remain inaccessible to many patients.

1.4 | Development of this position paper

The Federation of International Societies of Pediatric Gastroenterology, Hepatology, and Nutrition (FISPGHAN) Council identified the topic of infantile cholestatic liver disease as a global priority to be addressed by an expert team consisting of pediatric hepatologists nominated by the FISPGHAN individual member societies.

The expert team collaborated through teleconferences and digital communication from January 2023 to July 2024. Through a process of proposals, discussions, and voting, the team reached an agreed platform to produce this position paper, focusing on three main subtopics: (1) general diagnostic evaluation, (2) genetic testing, and (3) BA. The content includes new diagnostic modalities and future perspectives. Specifically, this article adopts a global perspective, considering the different needs of countries with limited resources. It aims to broadly serve as an informative resource for those involved in the care of infants with cholestatic liver disease.

Meanwhile, a comprehensive approach to neonatal cholestasis has been described in detail by NASPGHAN and ESPGHAN.[3] Further building on this guideline, the current FISPGHAN position paper intends to expand the perspective globally and incorporates the updated knowledge of diagnostic approaches for infantile cholestasis,

What is known

- Cholestasis, defined as impaired bile flow, is the most common manifestation of hepatobiliary disease in infancy and may progress to liver cirrhosis or failure.
- All neonates with a serum direct bilirubin ≥1 mg/dL (17 mmol/L) need to be evaluated for etiologies of cholestatic liver disease, especially for timely diagnosis of biliary atresia and treatable diseases.
- The etiology of extrahepatic and intrahepatic cholestasis is highly diverse, encompassing infections, genetic intrahepatic cholestasis and metabolic disorders, developmental defects, endocrine and immune-mediated conditions, as well as toxic and environmental factors.

What is new

- Universal screening for biliary atresia using either stool color cards or direct bilirubin testing should be implemented based on local expertise and available resources to facilitate earlier diagnosis of treatable conditions.
- Combining multiple diagnostic modalities can assist in establishing indications for intraoperative cholangiography and possible surgery for biliary atresia. Referral centers are encouraged to develop individual algorithms to optimize use of resources.
- Advances in diagnostic testing, particularly next-generation sequencing (including gene panels, whole exome sequencing, and whole genome sequencing), have significantly enhanced the evaluation and management of patients. The referral centers should establish close links with regional genetic centers to provide the mutational analysis.

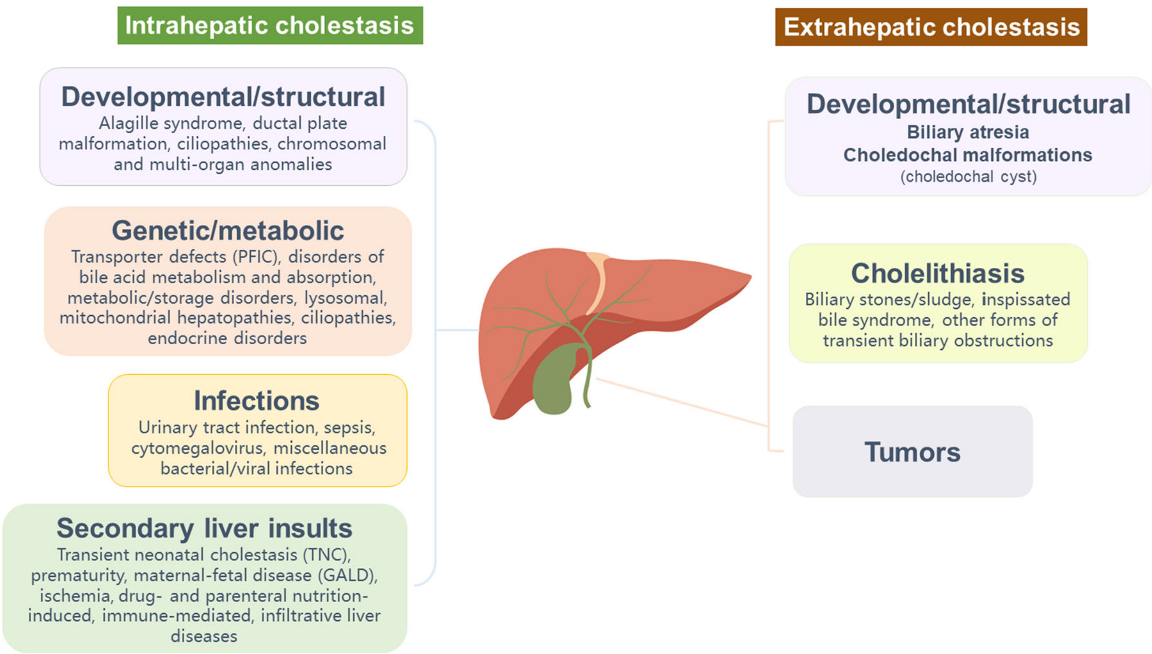

FIGURE 1 Summary of disease categories that may cause cholestasis in infancy. The diagnostic approach for infants with cholestasis relies on comprehensive and rational evaluations to identify the underlying pathophysiology causing the common symptoms and signs. An expanding list of disease entities is being identified with advanced genetic testing techniques with cautious interpretation based on both phenotype and genotype. GALD, gestational alloimmune liver disease; PFIC, progressive familial intrahepatic cholestasis. (Please refer to Tables 3 and S1.)

with a particular focus on BA and genetic intrahepatic cholestasis and metabolic disorders.

A comprehensive literature search was conducted in PubMed, MEDLINE, EMBASE, and the Cochrane Database of Systematic Reviews for publications in the English language through June 30, 2023, with additional references published ~2024 added during the revision. The search terms primarily focused on cholestasis and related diseases in infancy, including infantile cholestasis, neonatal cholestasis, BA, neonatal hepatitis, genetic intrahepatic cholestasis, metabolic disorders, metabolic liver disease, and neonatal infectious hepatitis. Cohort studies, cross-sectional studies, epidemiological studies, and consensus statements/guidelines published in the English language were included. No limitations were applied to the earliest search period, and the search concentrated on epidemiology, diagnosis, investigations, genetic testing, and BA.

Each author contributed to one of the three subtopics and drafted an initial version in the form of a narrative review based on the literature search. These versions of the full content were subsequently revised collectively through a consensus among all authors, producing the final manuscript and the formulation of clinical recommendations, which were reviewed and approved by all authors. The FISPGHAN council has reviewed this manuscript, provided comments from member society representatives, and, after minor modifications, approved it as a FISPGHAN position paper. All the authors have approved these additional modifications to the final version.

2 | UNIVERSAL SCREENING AND TARGETED SCREENING

2.1 | Universal screening

Two main strategies are used for the universal screening of neonatal cholestasis: stool color cards and direct bilirubin screening. Strategies involving validated neonatal stool cards distributed to all parents after birth, as well as to the first-line healthcare personnel, have been employed in many countries, including Japan, Taiwan, Switzerland, South Korea, Canada, and Argentina. These strategies have resulted in earlier referrals and improved native liver survival due to earlier BA diagnosis and treatment.[9,28–30] Another highly sensitive approach in the United States of America utilized the conjugated or direct bilirubin fraction on neonatal blood spots as early screening for cholestatic conditions, including BA.[31–33] According to Harpavat et al., a two-stage screening process was implemented. In the first stage, conducted within 60 h after birth, the upper limit of normal for direct bilirubin was determined at each site using the 97.5th percentile value, whereas the cutoff for conjugated bilirubin was set at 0.2 mg/dL. In the second stage, performed at 2 weeks of life, infants were considered to have negative screening results if their repeat direct or conjugated bilirubin levels were less than or equal to their Stage 1 levels and 1 mg/dL or less, as per NASPGHAN/ESPGHAN recommendations.[31] The universal screening requires a clear and firm commitment and

TABLE 1 Initial evaluation of an infant with cholestatic jaundice.

History	Possible underlying cause/diagnosis
Antenatal history	
In utero growth restriction/maternal infection	Maternal infections, such as CMV, syphilis, toxoplasma, and rubella, ALGS
Antenatal ultrasonography findings	Choledochal malformation, cholelithiasis, bowel anomalies, laterality defects/ BASM, cystic form of BA, CPSS, and cystic fibrosis
Intrahepatic cholestasis of pregnancy	Maternal heterozygous status for a pathogenic variant in a gene associated with PFIC (most commonly *ABCB4* or *ABCB11*).
Family history	
History of cholestasis in parents, siblings, or other family members;	AR disorders (cystic fibrosis, PFIC, α-1-antitrypsin deficiency, etc.)
Consanguinity	AR and AD disorders (ALGS)
Infant history	
Gestational age, SGA, perinatal distress	TNC, neonatal hepatitis,[a] congenital infection, congenital anomalies (including BASM), and genetic liver diseases
Hypoxia	Ischemic hepatitis
Neonatal infection and sepsis	Sepsis-related cholestasis, urinary tract infection, CMV, HIV, etc.
Diarrhea	PFIC1, *MYO5B* deficiency, mitochondrial hepatopathies, galactosemia, and hereditary fructose intolerance
Hemolytic disorders, use of antibiotics	Inspissated bile syndrome and cholelithiasis
Positive routine newborn screening	Cystic fibrosis, galactosemia, and NICCD
Parenteral nutrition ≥2 weeks	Parenteral nutrition-associated liver disease and TNC
Poor feeding: breastmilk- or formula-fed	Galactosemia and sepsis
Delayed passage of meconium	Cystic fibrosis
Physical examination	**Possible underlying cause/diagnosis**
General condition	
Unwell and vomiting/poor feeding	Urinary tract infection, metabolic disorders including galactosemia, sepsis, panhypopituitarism, and hypothyroidism
Petechiae/excessive bleeding	CMV infection, bacterial infection, coagulopathy, vitamin K deficiency, and liver failure
Acholic stool, dark urine	Biliary obstruction, Intrahepatic cholestasis with markedly reduced bile flow
Abnormal odors	Metabolic diseases (tyrosinemia: boiled cabbage; hyperammonemia: fetor hepaticus), bacterial infection
Head and facial appearance	
Microcephaly	CMV infection, trisomy 21
Dysmorphic facies	Chromosomal disorders (trisomies), Zellweger syndrome, ALGS (triangular facies, prominent forehead, hypertelorism, deep-set eyes; some infants may develop these typical facial features later)
Hearing problems	Specific inherited diseases (may present later), congenital infections (e.g., CMV), and FIC1 defects
Eye examination	
Cataracts	Rubella and galactosemia
Chorioretinitis	Congenital infections
Nystagmus and hypoplasia of the optic nerve	Septo-optic dysplasia/panhypopituitarism and mitochondrial disorders
Posterior embryotoxon (slit lamp exam)	ALGS

Physical examination	Possible underlying cause/diagnosis
Cardiac	
Murmur and heart failure	ALGS, BA (syndromic type), and cardiac diseases associated with specific metabolic disorders
Abdomen	
Ascites, prominent abdominal wall veins, enlarged, firm liver, and splenomegaly	Advanced chronic liver disease

Note: Modified from Fawaz et al.[3]

Abbreviations: AD, autosomal dominant; AFLP, acute fatty liver of pregnancy; ALGS, Alagille syndrome; AR, autosomal recessive; BA, biliary atresia; BASM, biliary atresia splenic malformation; CMV, cytomegalovirus; CPSS, congenital portosystemic shunt; HIV, human immunodeficiency virus; NICCD, neonatal intrahepatic cholestasis caused by citrin deficiency; PFIC, progressive familial intrahepatic cholestasis; SGA, small for gestational age; TNC, transient neonatal cholestasis; USS, ultrasonography.

[a]Neonatal hepatitis is an old term, referring to the histopathological findings, and may be seen in a variety of disorders like BSEP, CMV, and metabolic disorders. Neonatal hepatitis is now reserved for liver inflammation caused by identifiable pathogens.

support from the health authorities to ensure its optimal implementation. In the implemented large-scale screening programs, BA and other cholestatic diseases were diagnosed earlier.[9,31]

2.2 | Targeted screening

It is mandatory to measure total and fractionated direct (conjugated) bilirubin for symptomatic infants, or the asymptomatic ones who screen positive using any screening methods. Additionally, all infants with jaundice persisting beyond 2 weeks of age, regardless of the type of feeding, or those with pale/acholic stools at any age, with enlarged liver or spleen, should be tested for fractionated bilirubin (Tables 1 and 2). Fractionated bilirubin can be measured as either conjugated bilirubin or direct bilirubin. Specific measurement of conjugated bilirubin is preferred when available.[3] Caution should be taken in centers that use direct bilirubin (diazo method) levels (>17 μmol/L) as a cutoff, as this may result in a high rate of false-positive interpretations in conditions of elevated indirect hyperbilirubinemia.

Global perspective

- Two types of universal screening for newborns have been developed and implemented on a large scale: the stool color card and the direct or conjugated bilirubin measurement.
- Both tests have been shown to be effective in the earlier diagnosis of BA and the timely performance of KPE.
- A clear protocol for the recommended healthcare policies, coupled with infrastructure support from the health authorities and medical institutions, is mandatory to ensure optimal implementation.

3 | DIAGNOSTIC INVESTIGATIONS

3.1 | General evaluation

If the infant has direct bilirubin ≥1 mg/dL (17 μmol/L), a detailed history and thorough physical examination are essential. The priority is to focus on the timely identification of BA through a careful medical history, searching for "red flags," and performing laboratory, imaging, and pathological examinations for other causes (Tables 1 and 2). Next, the evaluation should be tailored to the gestational and postnatal age, onset of cholestasis, as well as the clinical assessment, to identify treatable medical conditions (e.g., urinary tract infection, hypothyroidism, hypopituitarism, galactosemia, NICCD, and tyrosinemia).[3,7,15,18,51–55]

Two tiers of laboratory evaluation and imaging studies are recommended for cholestatic infants (Table 2). Tier 1 includes full blood count, liver and coagulation profiles, urine analysis, and fasting ultrasound, all of which are widely available and should be considered mandatory. Tier 2 tests are used to differentiate specific etiologies within broader diagnostic categories.

Two key tests within Tier 1 recommendations are GGT and fasting abdominal ultrasound. Serum GGT is a more specific liver marker for cholestasis compared to AST and ALT, although it is also found in a variety of other tissues (e.g., placenta) and has different normal values in the newborns, particularly the premature ones. A mildly to moderately raised GGT value is commonly associated with hepatobiliary diseases, with markedly increased levels for age in obstructive cholestasis such as BA, Alagille syndrome, or *ABCB4* disease. In contrast, a normal or low serum GGT is found in several types of progressive familial

TABLE 2 Laboratory and imaging evaluation of the patients investigated for cholestatic liver disease in infancy.

Tier 1: Initial laboratory investigations and imaging studies		
Lab test	**Rationale**	**Specific disorders**
Total and direct bilirubin	Cholestasis is defined as a serum direct bilirubin greater than 1 mg/dL (17 μmol/L).	Extrahepatic and intrahepatic cholestasis
AST and ALT	Elevated in various forms of hepatocyte injury. ALT is more liver-specific.	May also be elevated in the injuries to skeletal muscles, heart, and kidneys (AST/ALT), as well as mitochondrial injuries and red blood cell diseases (AST).
Serum GGT[a]	Specific to hepatobiliary diseases, but is also found in a variety of tissues such as the placenta, and is physiologically elevated in premature babies and newborns.	Disproportionally low in most of the PFICs except MDR3 deficiency (also referred to as PFIC3).
Albumin	Reflects liver synthetic function	Reduced in cirrhosis/chronic liver failure
Glucose	Liver metabolic functions involve gluconeogenesis and glycolysis.	Low in galactosemia, mitochondrial disorders, panhypopituitarism, TNC associated with prematurity, and GALD.
Full blood count (complete blood count)	Infections and hematologic causes of jaundice	– Anemia: specific disorders: hemolysis (inspissated bile duct syndrome, galactosemia) – Leukocytosis: specific disorders: sepsis-related cholestasis – Thrombocytopenia: specific disorders: sepsis-related cholestasis and hypersplenism
Coagulation profile	Vitamin K deficiency due to insufficient bile flow and fat-soluble vitamin malabsorption, defects in the liver synthetic function	Severe cholestatic liver disease, decompensated cirrhosis, liver failure, and mitochondrial liver diseases
Blood cultures	Detecting bacteria or fungi in the blood	Bacteremia or sepsis
Urine microscopy and culture	Detecting bacteria or fungi in the urine	Urinary tract infection
Urine for nonglucose reducing substance[b]	Indicates the presence of galactose or fructose in urine	Galactosemia, NICCD, and hereditary fructose intolerance
Endocrine tests: TSH, thyroxine, and cortisol	May secondarily cause jaundice/cholestasis (hypothyroidism is included in newborn screening in many countries, with testing of TSH or thyroxine levels)	Panhypopituitarism and hypothyroidism
Abdominal ultrasonography[c]	Detecting dilated bile ducts, focal lesions/tumors, assessing the size of pre- and postprandial gallbladder, triangular cord sign, and splenomegaly.	Choledochal malformations, liver tumors, and other hepatobiliary anomalies Gallbladder appearances and size and triangular cord sign may point to BA, but have insufficient sensitivity and cannot exclude BA when absent.[a]

Tier 2: If specific etiology is considered or after consultation with a pediatric hepatologist		
Lab test	**Rationale**	**Specific disorders**
Serology for congenital infections	Depending on local epidemiology	CMV is the most important cause
Sweat chloride analysis (or serum immunoreactive trypsinogen level or CFTR mutation analysis, where appropriate)		Cystic fibrosis
α-1-antitrypsin phenotype or genotype		α-1-antitrpysin deficiency
Total serum bile acids	Disproportionately low in primary bile acid synthesis defects; High in other causes of cholestasis	Inborn errors of bile acid metabolism
Mass spectrometry for urinary bile acid analysis[d]	Confirmatory diagnosis of inborn errors of bile acid metabolism and to specifically identify abnormal bile acid metabolites	Inborn errors of bile acid metabolism

TABLE 2 (Continued)

Tier 2: If specific etiology is considered or after consultation with a pediatric hepatologist		
Lab test	**Rationale**	**Specific disorders**
Survey for metabolic disorders: ammonia, lactic acid, blood sugar, tandem mass spectrometry,[b] plasma amino acids, urine succinyl acetone, plasma acylcarnitine, creatine kinase	To screen or diagnose miscellaneous metabolic disorders, including urea cycle disorders, galactosemia, fatty acid oxidation defects, mitochondrial hepatopathies, tyrosinemia type I, when clinical suspicion, based on neurological symptoms, growth and developmental delay, or organomegaly, is present	Genetic and metabolic disorders
Ferritin	Suspicion of gestational alloimmune disorder is elevated with compatible clinical features (e.g., neonatal liver failure and repeated fetal loss)	GALD
Alpha-fetoprotein	Elevated in tyrosinemia type 1 and in different forms of hepatocellular injury in early infancy	Tyrosinemia, neonatal hepatitis, TNC, GALD, and liver tumors
Imaging	**Rationale**	**Specific disorders**
MRCP, hepatobiliary scintigraphy, ERCP, and PTCC	Variable specificity and sensitivity in diagnosing BA	BA and choledochal malformation
Chest and skeletal x-ray	Exclude lung infections, look for butterfly vertebrae, and signs of rickets	ALGS and vitamin D deficiency in chronic cholestasis
Ultrasonography (for evaluation of involved organs)	If ALGS, syndromic BA, or multisystemic disorders are suspected	ALGS, BASM, and congenital anomalies
Emerging new diagnostic modalities[e]	**Rationale**	**Specific disorders**
MMP-7	Serum levels are significantly elevated in patients with BA compared to those with other causes of neonatal cholestasis	BA and other causes of cholestasis
Transient elastography or shear wave elastography	Measure liver stiffness	Can be used in the differential diagnosis of BA and for assessing the severity of liver fibrosis
Consultation	**Rationale**	**Specific disorders**
Ophthalmologist	Eye assessment for posterior embryotoxon and chorioretinitis	ALGS and congenital infections
Hepatologist or interventional radiologist	Consider percutaneous liver biopsy	Wide spectrum of hepatobiliary disorders
Geneticist	If genetic etiologies of cholestasis are suspected, or for discussion on interpretation of genetic test results (see Table 3)	Genetic and metabolic diseases

Abbreviations: ALGS, Alagille syndrome; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BA, biliary atresia; BASM, biliary atresia splenic malformation; CFTR, cystic fibrosis transmembrane receptor; ERCP, endoscopic retrograde cholangiopancreatography; GALD, gestational alloimmune liver disease (previously referred to as neonatal hemochromatosis); GCMS, gas chromatography-mass spectrometry; GGT, gamma-glutamyl transpeptidase; KPE, Kasai portoenterostomy; MMP-7, matrix metalloproteinase-7; MRCP, magnetic resonance cholangiopancreatography; PFIC, progressive familial intrahepatic cholestasis; PTCC, percutaneous transhepatic cholecysto-cholangiography; TMS, tandem mass spectrometry; TSH, thyroid-stimulating hormone.

[a]GGT levels: A GGT level greater than approximately 200–300 U/L is suggestive of BA, although lower levels may also be seen in some cases.[34,35] In patients with intrahepatic cholestasis, low GGT levels may suggest conditions such as PFIC, while high GGT levels may indicate ALGS or certain types of PFIC, warranting further diagnostic evaluation such as genetic testing.[22,36]

[b]Urine-reducing substances can be used for selecting candidates for further confirmatory diagnostic tools in resource-limited areas. If tandem mass spectrometry or genetic analysis is readily available, the physician should proceed directly to these tests. TMS or GCMS should be performed in selected cholestatic infants with suspected fatty acid oxidation defects, urea cycle defects (including citrullinemia), or tyrosinemia, and should be used judiciously in resource-limited settings.

[c]Ultrasonography: Typical findings suggesting BA include gallbladder abnormalities such as non-visualization (atretic) or a small (ghost) gallbladder <1.5–2 cm, abnormal gallbladder wall and shape, and lack of contractility after feeding; the presence of the triangular cord sign; or enlarged hepatic artery. Gallbladder should be assessed after 3–4 h of fasting to allow the gallbladder to fill. A longer fasting time requires intravenous fluids for infants to avoid hypoglycemia or dehydration. Gallbladder abnormalities and the triangular cord sign may suggest BA, but their absence does not exclude it (highly specific but only moderately sensitive). The sensitivity and specificity of ultrasound findings for the diagnosis of BA have been summarized in meta-analyses and reviews.[34,35,37,38]

[d]Primary bile acid synthesis defects (inborn errors of bile acid metabolism) are important causes of cholestasis that are treatable if diagnosed early.

[e]MMP-7, transient elastography (FibroScan), and shear wave elastography are emerging diagnostic modalities with promising value in differentiating BA from other cholestatic conditions, as well as for assessing liver fibrosis after KPE. Cutoff levels vary depending on the modality used and the clinical context—such as screening, diagnosis, or evaluation of disease severity and progression.[34,35,39–50]

intrahepatic cholestasis (PFIC), primary bile acid synthesis defects (with low serum bile acid levels), and arthrogryposis-renal dysfunction-cholestasis syndrome (ARC) syndrome. A disproportionately low or high GGT is an indication for further, more specific investigations.[7,19,22,36,51–58]

Fasting abdominal ultrasound is a noninvasive diagnostic modality that is particularly helpful in the evaluation of gross anatomic abnormalities such as choledochal malformations.[59] Specific findings have been reported to support the diagnosis of BA, but they require a greater operator expertise. Typical findings suggesting BA includes gallbladder abnormalities: non-visualization (atretic) or small (ghost) gallbladder, abnormal gallbladder wall and shape, lack of contractility after feeding; or the presence of "triangular cord" sign.[34,37,38] (Table 2) Similar diagnostic value is attributed to situs inversus abdominis with the presence of various vascular and splenic malformations.[59,60] However, it is of note that an unremarkable ultrasound examination does not exclude BA.[38]

Multiple imaging modalities, including hepatobiliary scintigraphy, magnetic resonance cholangiopancreatography, and endoscopic retrograde cholangiopancreatography (ERCP), have been used in neonatal cholestasis. Their specificity and sensitivity in diagnosing BA are variable depending on local institutional experiences.[59–63] Percutaneous transhepatic cholecysto-cholangiography (PTCC) has been described in infants with a normal gallbladder; the visualization of the biliary tree can exclude BA and may avoid an unnecessary intraoperative cholangiogram in infants where BA is still suspected.[64,65] However, these diagnostic modalities have not been universally adopted due to considerable variability in the positive predictive value for BA (scintigraphy),[66] need for endoscopic expertise (ERCP) or interventional radiologist expertise (PTCC), and local differences in the health system accessibility.

Some emerging biomarkers and diagnostic modalities have been reported to be promising in distinguishing BA from other forms of neonatal cholestasis such as serum levels of matrix metalloproteinase-7 (MMP-7), or ultrasound-based elastography including transient elastography (Fibroscan) or shear-wave elastography, which assess the extent of liver fibrosis.[34,35,39–50] Given that different cutoff values have been developed for various clinical situations, such as screening, diagnosis, and the evaluation of disease severity or progression, it is important to be cautious about the purpose for which these parameters are used. These observations require prospective use and validation to be considered for a wider application.

Percutaneous liver biopsy provides diagnostic and prognostic information on the liver condition, but must be correlated with the clinical background and laboratory investigations. Liver histopathology is valuable in diagnosing both extrahepatic and intrahepatic cholestasis, including conditions such as biliary obstruction, chronic inflammatory changes, metabolic disorders, including lipid storage diseases, mitochondrial disorders, infiltrative hematological diseases, drug-induced liver injuries, etc. The pathological features of BA include loose, edematous portal tracts with fibrosis; bile ductular proliferation; and cholangiolar bile plugs. In clinically confirmed cases of BA, pathologists' diagnostic accuracy for detecting obstruction ranged from 79% to 98%, with a positive predictive value of 91%.[67,68] Additional immunostaining and electron microscopic examinations could assist in investigating cryptogenic or emerging genetic diseases.[19,56,60,69,70] It should be noted that considerable expertise in the interpretation of liver pathology is needed, as conditions such as Alagille syndrome, α-1-antitrypsin deficiency, cystic fibrosis, PN-associated cholestasis, and some types of PFICs can mimic the histologic features of BA, especially when the tissue sample is limited.[53] Giant cell transformation (giant cell hepatitis) or extramedullary hematopoiesis can be seen in various diseases such as CMV hepatitis, BSEP deficiency, or inborn errors of bile acid synthesis. Other specific histopathological findings include paucity of intrahepatic bile ducts in Alagille syndrome, features of ductal plate malformation in fibrocystic liver disease (including polycystic diseases), and characteristic storage substances in inherited lysosomal disorders. In genetically confirmed cholestatic conditions, liver histopathology is optional and depends on the available facilities, parents' preferences, and risk-benefit considerations. The liver biopsy remains indispensable in understanding the pathogenesis, assessing the severity of liver disease, and diagnosing specific diseases. On the other hand, its future role in the diagnostic algorithm will likely diminish with the integration of emerging noninvasive diagnostic modalities, including rapid genetic tests, ultrasound-based elastography, and novel biological markers.

The intraoperative cholangiogram (IOC) remains the gold standard for diagnosing BA. A concomitant liver biopsy is recommended to assess liver disease severity in BA and to diagnose other conditions. It should be noted that IOC should assess the patency of biliary tree, both proximal and distal to the gallbladder. If the IOC shows a patent biliary system, further investigations are needed to determine other etiologies once BA has been excluded, such as Alagille syndrome or conditions causing diminished bile flow and pale stools such as PFIC, alpha-1 antitrypsin deficiency or other forms of intrahepatic cholestasis.[35,64]

Recently, a laparoscopic cholangiography and liver biopsy with subsequent Kasai operation during the same procedure has been described.[71]

Global perspective

- The first step is to diagnose BA as early as possible to ensure timely KPE.
- The "red flags" for BA or genetic and metabolic diseases are: persistent jaundice beyond 2 weeks, dark urine, acholic/pale stools, deranged clotting, poor growth, vomiting/lethargy, and abnormal neurological signs.
- Initial evaluation of cholestatic liver disease can be performed at the primary care level in close collaboration with a referral center, aiming to diagnose or exclude BA as early as possible.
- The diagnostic modalities commonly used include clinical history, physical examination, Tier 1 and Tier 2 laboratory tests, emerging biomarkers, imaging studies based on the local access and expertise, and liver histopathology (Tables 1 and 2).
- Ultrasonographic exams, which are widely available, can be used to identify possible cases of BA by detecting the triangular cord sign and gallbladder abnormalities; however, negative findings do not exclude BA.
- Multiple diagnostic modalities should be promptly used to determine whether a patient should undergo intraoperative cholangiography and possibly require KPE.
- The liver biopsy remains indispensable in understanding the pathogenesis, assessing the severity of liver injury, and diagnosing specific diseases.

More in-depth diagnostic evaluation (Tier 2 tests) should be guided by the following: (a) medical history and physical examination, initial laboratory investigations, and imaging; (b) availability of local resources; (c) etiological spectrum of infantile cholestasis in the local setting.[3] A degree of flexibility and diversity can be adopted depending on the availability of local resources and expertise, as well as regional differences in the prevalence of specific etiologies.

Given the diverse diagnostic modalities, institutions can develop algorithms to optimize resources, minimize the time to diagnose or exclude BA, and train personnel with adequate expertise for conducting and interpreting each examination. Education of medical personnel and the public is encouraged to be updated with the progress in understanding and management of cholestatic liver disease. An algorithm for the evaluation of cholestatic liver disease is shown in Figure 2.

4 | GENETIC TESTING: IMPACT ON THE EVALUATION AND MANAGEMENT OF CHOLESTASIS

Advances in the NGS techniques, lower cost, and faster turnaround times have enabled the integration of genetic testing into the general diagnostic evaluation of cholestasis in infancy. Furthermore, the development and increased accessibility of genetic testing have led to the identification of numerous novel genetic variants. Currently, genetic etiologies for neonatal cholestasis comprise ~25%–50% of all causes, with considerable variation depending on the target population under investigation.[3,19,21,22,72–74] The most common genetic etiologies of neonatal cholestasis and the associated disease phenotype assigned in the Online Mendelian Inheritance in Man (OMIM) database for PFIC subtypes are provided in Table 3.

Established panels that detect single gene variants in up to 60+ genes are set toward identifying Alagille syndrome, biliary transport defects, particularly PFIC, bile acid synthesis defects, alpha-1-antitrypsin deficiency, metabolic disorders (e.g., NICCD, Niemann–Pick disease, and galactosemia), mitochondrial disorders, or additional biliary disorders such as cystic fibrosis or genetic causes of neonatal sclerosing cholangitis.[22,53,72,75] Specific diseases vary in incidence in different regions but overall the most commonly implicated genes in cholestatic infants include *JAG1* and *NOTCH2* (Alagille syndrome), *SERPINA1* (alpha-1-antitrypsin deficiency), *ABCB11* (BSEP deficiency, also known as PFIC2), *ABCB4 (*MDR3 deficiency, also known as PFIC3*), ABCC2* (Dubin–Johnson syndrome), and *SLC25A13* (citrin deficiency) (Table 3).[22,74–77]

Genetic testing can be very useful in discriminating inherited conditions from other causes of neonatal cholestasis. Given the urgency in excluding BA and diagnosing treatable genetic intrahepatic cholestasis and metabolic disorders, the confirmation of a patent biliary-enteric connection and genetic testing can be performed either concurrently or sequentially, After excluding BA, genetic testing using NGS techniques can achieve a diagnosis in 10%–30% of neonates and infants with cholestasis depending on the stringency of the patient selection.[20–22,74,75,77]

Approaches utilizing different genetic sequencing tools, including Sanger sequencing, NGS panels, or whole-exome sequencing (WES), have been applied in clinical settings. The choice of genetic testing tools depends on clinical indication, disease severity, urgency, and available resources.[73–75,77] Genetic panels offer the advantage of simultaneous screening for multiple disease-causing genetic variants that may have overlapping clinical features and low individual prevalence. WES can be performed by assessing only an index case and interpreted based on correlation with phenotype and a database of known disease-causing variants of genetic diseases, or simultaneously testing trio sequencing (including the index case and parents) that can incorporate the inheritance pattern into interpretation. WGS has transitioned from research into clinical practice and is expected to become accessible in the coming years due to the continuous decrease in cost.[78] The use of trio sequencing, whether employing WES or WGS, to validate de novo variants associated with diseases or to reveal new genetic disorders, is highly valuable.

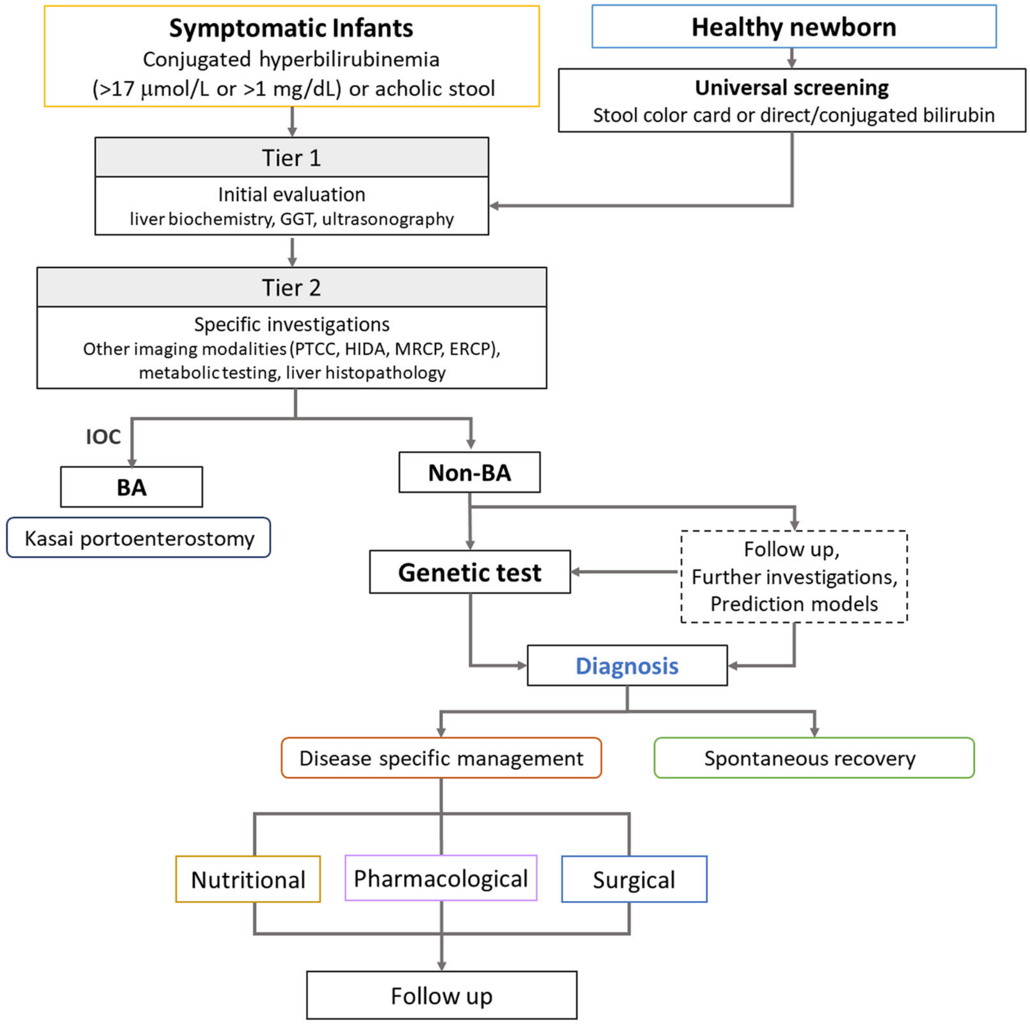

FIGURE 2 Concept and flow chart of diagnosis and management of infants with cholestatic liver diseases. ERCP, endoscopic retrograde cholangiopancreatography; GGT, gamma-glutamyl transferase; HIDA, hepatobiliary iminodiacetic acid scan; IOC, intraoperative cholangiography; MRCP, magnetic resonance cholangiopancreatography; PTCC, percutaneous transhepatic cholecysto-cholangiography.

Expansion of genetic testing has also led to the identification of an association between heterozygosity of mutations in specific genes and TNC, particularly *ATP8B1*, *ABCB11*, and *ABCB4*.[79–82] However, the precise relationship between individual heterozygous mutations and severity of clinical phenotype in patients with TNC has not been established and complicates genetic counseling.[76,83] Furthermore, it is important to acknowledge that as more data is accumulated, mutations previously classified as benign may become pathogenic and vice versa, resulting in the identification of new disease-causing variants.[83] Consequently, repeat testing or reanalysis may be indicated in patients with chronic or progressive disease with negative genetic results at the initial evaluation. Dissemination of results from genetic testing is critically important to better define associations with phenotype. However, the genetic results should be interpreted with caution, especially in situations involving insufficient evidence of disease-causing variants or variants from multiple genes, since the diagnosis of a genetic disorder can significantly negatively impact patients and their families.

Achieving an accurate genetic diagnosis enables: (a) early identification of treatable diseases such as bile acid synthetic defects or NICCD, or candidates for recently developed drugs such as ileal bile acid transporter inhibitors for Alagille syndrome and PFIC,[84,85] (b) predict prognosis, (c) modify medical management based on prognosis emphasizing nutritional care, prevention of cirrhosis-related complications, consideration of liver transplantation, and (d) family counseling and prenatal diagnosis for the future pregnancies. A collaborative team including a geneticist, pediatric gastroenterologist/hepatologist, and bioinformatic specialist should be established to correlate genetic and clinical phenotypic data and to

TABLE 3 An overview of major genes associated with cholestatic liver disease or direct hyperbilirubinemia in infancy.

Disease category	Gene(s)	Disease name(s)	Mode of inheritance
Abnormal bile duct development	*JAG1*	ALGS	AD
	NOTCH2	ALGS	AD
PFIC,[a] transporter defect, genes involving abnormal cell structures	*ATP8B1*	FIC1 deficiency	AR
	ABCB11	BSEP deficiency	AR
	ABCB4	MDR3 deficiency	AR
	TJP2	TJP deficiency	AR
	NR1H4	FXR deficiency	AR
	SLC51A	OST-alpha deficiency	AR
	USP53 KIF12		AR
	ZFYVE19		AR
	MYO5B		AR
Disorders of bile acid metabolism and absorption (inborn errors of bile acid metabolism, bile acid synthesis, or conjugation disorders)	*AKR1D1*	BASD	AR
	ACOX2	BASD	AR
	AMACR	BASD	AR
	CYP7B1	BASD	AR
	HSD3B7	BASD	AR
	CYP27A1	Cerebrotendinous xanthomatosis	AR
	BAAT	Bile acid conjugation defect	AR
	SLC27A5	Bile acid conjugation defect	AR
	SLC51B	OST-BETA defect - Primary bile acid malabsorption	AR
Etabolic disorders (carbohydrate, amino-acid, lipid metabolic/storage disorders)	*GALT*	Galactosemia	AR
	SLC25A13	NICCD	AR
	CFTR	Cystic fibrosis	AR
	FAH	Tyrosinemia	AR
	NPC1	Niemann–Pick Disease type C1	AR
	NPC2	Niemann–Pick Disease type C2	AR
	PEX genes	Peroxisome biogenesis disorders	AR
	SERPINA1	Alpha-1-antitrypsin deficiency	AR
	LSR	LSR deficiency	AR
	ALDOB	Hereditary fructose intolerance	AR
	LIPA	Lysosomal acid lipase deficiency: CESD and Wolman disease	AR
Mitochondrial disorders	*POLG*	Mitochondrial DNA depletion syndrome	AR
	MPV17		AR
	DGUOK		AR

(Continues)

Disease category	Gene(s)	Disease name(s)	Mode of inheritance
Ciliopathies	*DCDC2*	Neonatal sclerosing cholangitis	AR
			AR
	CLDN1	Ichthyosis, leukocyte vacuoles, alopecia, and sclerosing cholangitis	AR
	PKHD1	Autosomal recessive polycystic kidney disease	
	PKD1L1	BASM	AR
	Genes associated with NPHP	Nephronophthisis	AR
Abnormal organ development	*HNF1B*	Renal cysts and diabetes syndrome	AD
Disorders with multiple organ involvement	*VPS33B* or *VIPAR*	Arthrogryposis, renal dysfunction, and cholestasis syndrome	AR
	UNC45A	Aagenaes syndrome/lymphedema cholestasis syndrome 1	AR
	TTC37	Tricho-hepato-enteric syndrome	AR
Bilirubin metabolism defects[b]	*ABCC2* (MRP2 defect)	Dubin–Johnson syndrome	AR
	SLCO1B1/SLCO1B3	Rotor syndrome	DR

Abbreviations: ABCB11, ATP binding cassette subfamily B member 11; ABCB4, ATP binding cassette subfamily B member 4; ACOX2, acyl-CoA oxidase 2; AD, autosomal dominant; AKR1D1, aldo-keto reductase family 1 member D1; ALDOB, aldolase, fructose-bisphosphate B; AMACR, alpha-methylacyl-CoA racemase; AR, autosomal recessive; ATP8B1, ATPase phospholipid transporting 8B1; BAAT, bile acid-CoA: amino acid N-acyltransferase; BASD, bile acid synthetic defects; BASM, biliary atresia with splenic malformation; CESD, cholesteryl ester storage disease; CFTR, cystic fibrosis transmembrane conductance regulator; CLDN1, claudin 1; CYP27A1, cytochrome P450 family 27 subfamily A member 1; CYP7B1, cytochrome P450 family 7 subfamily B member 1; DCDC2, doublecortin domain containing 2; DGUOK, deoxyguanosine kinase; DR, digenetic recessive; FAH, fumarylacetoacetate hydrolase; GALT, galactose-1-phosphate uridylyltransferase; HSD3B7, hydroxy-delta-5-steroid dehydrogenase, 3 beta- and steroid delta-isomerase 7; JAG1, jagged canonical Notch ligand 1; KIF12, kinesin family member 12; LIPA, lipase A, lysosomal acid type; LSR, lipolysis stimulated lipoprotein receptor; MPV17, MPV17 mitochondrial inner membrane protein; MYO5B, myosin VB (myosin 5B); NICCD, neonatal intrahepatic cholestasis caused by citrin deficiency; NOTCH2, Notch receptor 2; NPC1, NPC intracellular cholesterol transporter 1; NPC2, NPC intracellular cholesterol transporter 2; NR1H4, nuclear receptor subfamily 1 group H member 4; OMIM, Online Mendelian Inheritance in Man; PEX genes, peroxisomal biogenesis factor genes (e.g., PEX1, PEX6—multiple members); PFIC, progressive familial intrahepatic cholestasis; PKD1L1, polycystin 1 like 1, transient receptor potential channel interacting; PKHD1, PKHD1 ciliary IPT domain containing fibrocystin/polyductin; POLG, DNA polymerase gamma, catalytic subunit; SERPINA1, serpin family A member 1; SLC25A13, solute carrier family 25 member 13; SLC27A5, solute carrier family 27 member 5; SLC51A, solute carrier family 51 subunit alpha; SLC51B, solute carrier family 51 subunit beta; TJP2, tight junction protein 2; USP53, ubiquitin specific peptidase 53; ZFYVE19, zinc finger FYVE-type containing 19.

[a]The OMIM database has designated up to 11 types of PFIC. Given the diverse genetic backgrounds and disease mechanisms involved, while many clinicians find the nomenclature of PFIC1 (ATP8B1 deficiency), PFIC2 (ABCB11 deficiency), and PFIC3 (MDR3 deficiency) useful in clinical communication, the expert team prefers referring directly to the specific genetic defect when discussing these particular genetic diseases.

[b]Although these genetic disorders causing hyperbilirubinemia do not lead to cholestasis, they remain important in the differential diagnosis when patients have direct-type hyperbilirubinemia above 1 mg/dL.

continue updating and revisiting them during patient follow-up if required (Table S1). Limitations of genetic testing include the availability of budget and resources, expertise required for data interpretation, and the necessity of effective communication of genetic information to primary care physicians and patient families. It is important to consider the authenticity and accuracy of the genetic reports, the psychological and social burden on a patient or family who undergoes the genetic testing process, the possibility of false-positive or false-negative results, and the associated ethical issues that might impact the patient and family, the healthcare system, and society.

Genetic testing will be more widely employed in the future, with long-term observational studies conducted to fully assess the impact of genetic testing on patient outcomes.

Global perspective

- For patients in whom the biliary patency is confirmed, investigations should be conducted with local flexibility and diverse approaches, including genetic tests.
- In resource-limited areas, patients prioritized for genetic testing should include those with low GGT, low blood glucose, or abnormal metabolic profiles, as well as those from consanguineous families or with multiple affected cases within a family.
- Patients with progressive liver disease, cirrhosis, or liver failure of unknown etiology, as well as those who are candidates for liver transplantation where BA is excluded, should also be considered for genetic testing.
- An accurate genetic diagnosis can lead to earlier, disease-specific management, help predict prognosis, optimize the treatment plan, and initiate family counseling or prenatal diagnosis for future pregnancies.

5 | BA REMAINS AN AREA OF UNMET NEEDS

BA is an idiopathic neonatal cholangiopathy characterized by progressive inflammatory obliteration of the extrahepatic and intrahepatic bile ducts. It is the most common indication for pediatric liver transplantation (LT) and, if left untreated, will progress to biliary cirrhosis, end-stage liver disease, and early death.[86] After KPE, approximately 50% of children with BA require LT before 2 years of age, and ultimately, around 75% require an LT by 20 years of age.[53,86–88]

Despite ongoing efforts to better understand its pathogenesis, BA remains an enigmatic chronic disease with limited modalities to modify its natural history.[87] Nevertheless, extensive clinical and basic research over the last half of a century has established some facts about BA. Earlier corrective surgery by KPE undoubtedly has a positive effect on native liver survival. A recent meta-analysis confirmed the importance of early diagnosis and surgery before 30 days of life in infants with BA on native liver survival at 5, 10, and 20 years.[88]

BA is a common clinical phenotype of several potentially overlapping pathogenic mechanisms. Several BA variants have been described, including some that may have an antenatal onset: (a) biliary atresia splenic malformation (BASM), associated with heterotaxy (laterality defects) and splenic malformations,[89] (b) non-BASM syndrome (associated with other congenital malformations),[90] and (c) cystic BA.[91] There are also variants with less certain timing of onset, such as (d) CMV-associated BA,[92] and (e) the most common—isolated or sporadic form.[93] Specific BA variants have been associated with different outcomes. For example, the outcome of patients with BASM may be affected by conditions such as fatal cardiac anomalies, as well as the age at which the KPE was performed.[89,92] Patients with cystic BA and patent proximal hepatic ducts can have more favorable outcomes,[94] while CMV-associated BA was reported to have a poorer response to KPE.[91]

Pathogenesis of BA is complex and poorly understood. It is driven by a strong pro-inflammatory response[95] possibly triggered by some extrinsic factors such as infections or even environmental chemical compounds and linked to immunologic factors (e.g., autoreactivity) or the host genetic predisposition.[96,97] Mendelian genetic factors do not appear to play a role in isolated BA.[79,87,89,97–101]

There is a lack of effective medical therapies for BA. Despite the well-documented upregulation of the pro-inflammatory immune response in both the blood and liver tissue,[95] clinical attempts to contain or suppress it with steroids, intravenous immune globulins, or rituximab have largely been unsuccessful.[100–102] BA patients with chronic cholestatic disease carry a high long-term medical burden, especially before LT. Their main clinical problems in adolescence and adulthood include portal hypertension and recurrent cholangitis.[103] Their education, social reintegration, and independence remain challenging, possibly due to a low-grade chronic encephalopathy that is often clinically unrecognized and warrants further validation.

Some strategies can be initiated before delivery to detect BA as early as possible. For example, a neonatal cholestasis card can be provided during pregnancy as a part of the Mother-Child Protection Card at the community level for the family and caregivers during the early neonatal period.[104] This could be followed by implementing universal screening after birth using either a stool color card or neonatal direct or conjugated bilirubin testing.[105] This practice needs to be evaluated in each local setting. Strategies such as artificial intelligence or smartphone applications for stool color detection and diagnostic algorithms are being developed and warrant further validation.

The major obstacle in advancing BA treatment remains a lack of understanding of etiopathogenesis. The geographic variation in the prevalence of BA and syndromic BA supports the presence of critical environmental and/or obscure genetic factors that, if identified, could lead to preventative management. One important unresolved question is why the prevalence of BA is higher in East Asia and French Polynesia (1–3/10,000 live births) in comparison to European and North American countries (~0.5/10,000 live births). Another mystery is why the prevalence of the syndromic forms is highly variable in different regions, with the highest of around 10%–15% in North America and Europe, and the lowest of <1%–3% in Asia and South America.[89,105–110] The prevalence may be lower with a strict definition of BASM and higher if any other congenital malformations were included.[107] In addition, multiple susceptibility genes have been investigated in BA, with only *PKD1L1* gene implicated so far to be related to a subset of patients with BASM.[98,99]

Another challenge lies in improving the management of patients after KPE. The corrective surgery alleviates only the extra-hepatic biliary obstruction and does not prevent the progression of intra-hepatic biliary injury. Many adjuvant medical therapies have been advocated to ameliorate intrahepatic bile duct damage, but only a few have been subjected to rigorous and objective assessment.[111,112] Customized use of post-KPE steroids and antibiotic therapy have been studied but require a challenging validation across larger patient cohorts.[100,112] Investigation into the utility of N-acetylcysteine in BA is ongoing, and early clinical results support its positive role in immune modulation.[113,114] Additional randomized clinical trials are assessing the efficacy of ileal bile acid transporter inhibitors.[115] Future therapies will integrate knowledge gained from emerging multi-omic studies to more selectively target unique aspects of the aberrant immune-driven pro-inflammatory injury in BA. The role of novel anti-inflammatory strategies should be assessed in the early post-KPE period, potentially suggesting more targeted treatments. A formal randomized trial of antiviral treatment should be considered to improve the outcome of CMV-associated BA. In addition, biomarkers and new tools for prognostic prediction have been developed, which may lead to more personalized medical care after KPE.[46,116]

Akin to many other rare conditions, clinical research into BA requires multicenter and international collaboration. Multicenter consortia such as CHiLDReN (the Childhood Liver Disease Research Network) in North America and BARD (Biliary Atresia and Related Disorders) in Europe have been bringing together the experience in this puzzling condition for over 20 years. Despite straightforward and well-established surgical management by KPE, difficulties still exist in a very individualized approach at different centers in some aspects of management, such as the use of steroids or antivirals, antibiotic prophylaxis, management of portal hypertension, and the timing of assessment for liver transplantation.

6 | CONCLUSIONS

There is an urgent need for more research into the epidemiology and etiologies of infantile cholestatic liver disease across different regions, particularly in resource-limited areas. With the advancement of diagnostic modalities and their cost-effectiveness, adopting a tailored diagnostic algorithm based on the local healthcare system and hospital setting should be encouraged. The fundamental elements, including medical history, physical examination, and focused laboratory investigations with fasting abdominal ultrasound, need to be prioritized and integrated into education programs for healthcare professionals. Seamless collaboration and referral systems with more specialized institutions are recommended. Ultimately, lower cost and broader dissemination of emerging diagnostic and treatment modalities will help improve outcomes for infants with neonatal cholestasis.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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